AIR QUALITY AND OZONE ANALYSIS ACROSS THE UNITED STATES (2020)

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Introduction

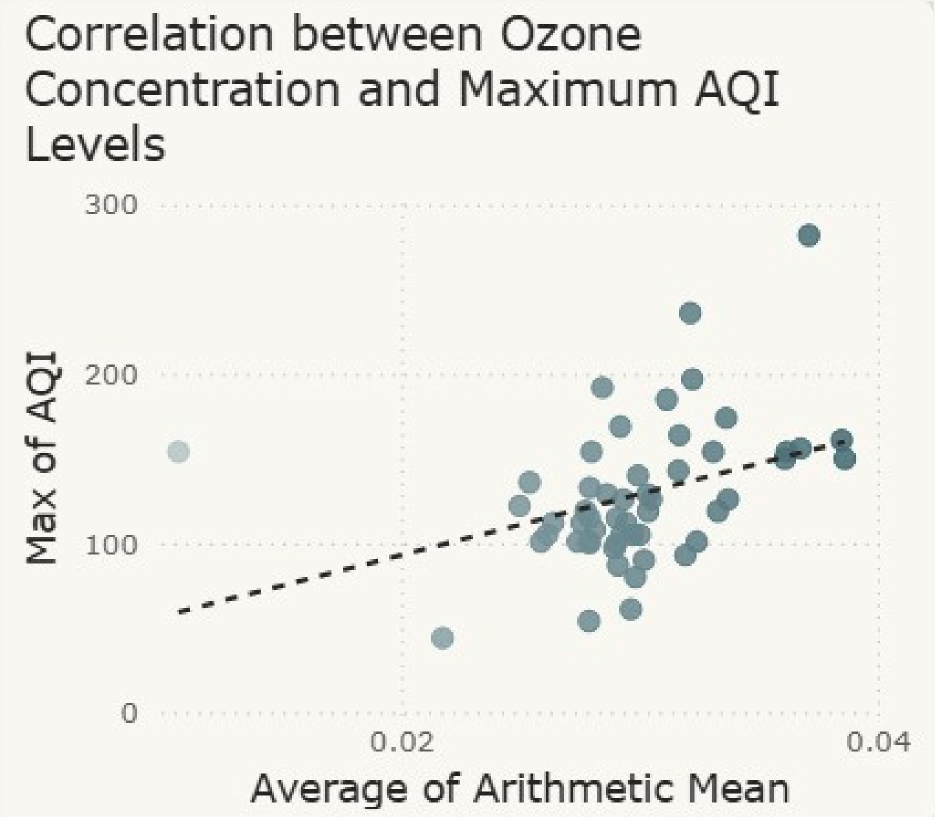
The dataset contains ozone and AQI data collected across the United States during 2020.
The data was gathered from outdoor monitoring stations.
The dataset includes air quality indicators, ozone concentration levels, states, counties, and time-based records.
The project aims to analyze air pollution patterns using Power BI visualizations.

Analysis Questions

- Is there a correlation between the average ozone concentration and the maximum AQI value?
- How are AQI levels geographically distributed across different states in the US?
- Which top 10 counties recorded the highest average AQI values?
- How did ozone pollution levels change across the months of 2020?
- Which states contributed the highest number of records in the dataset?

Is there a correlation between the average ozone concentration and the maximum AQI value?

VISUALIZATION

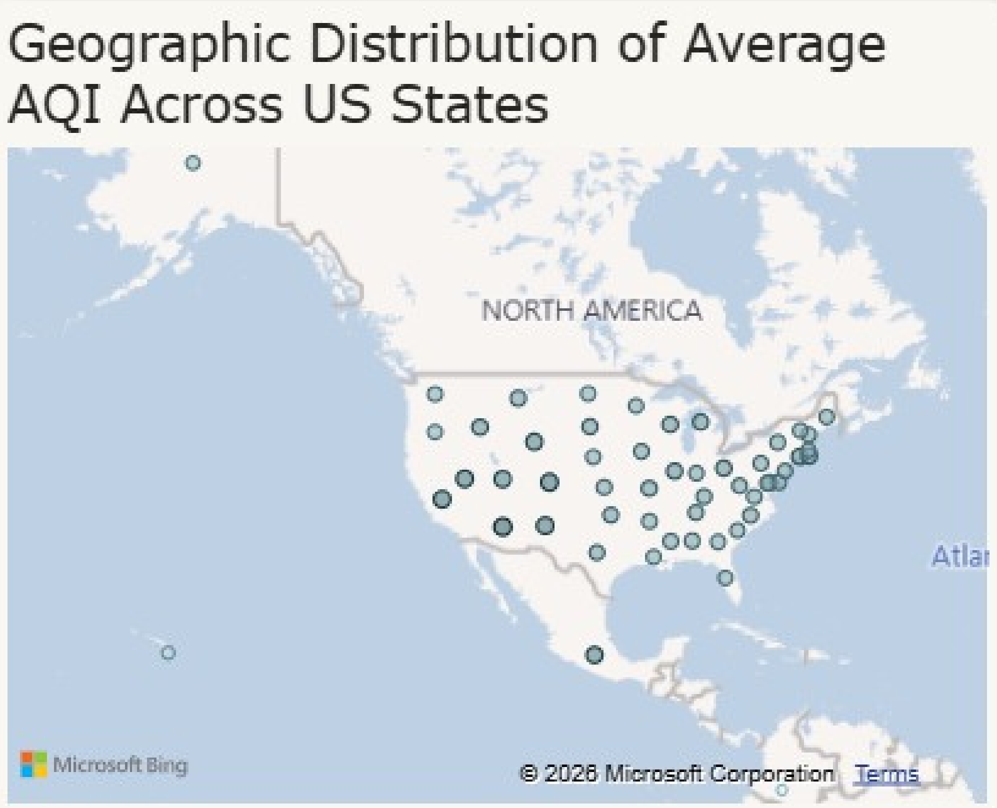


RESULTS

- A positive correlation can be observed between ozone concentration and AQI levels.
- Higher ozone concentration is generally associated with poorer air quality.
- The trend line indicates a moderate upward relationship between the two variables.

How are AQI levels geographically distributed across different states in the US?

VISUALIZATION

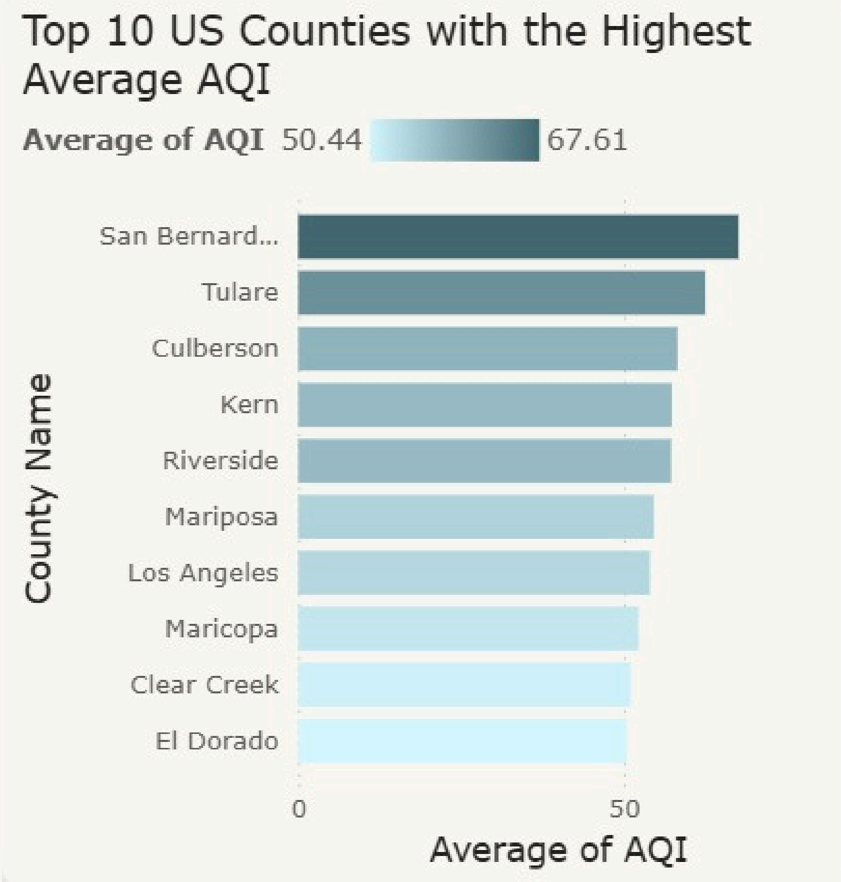


RESULTS

- The map shows that AQI measurements are distributed across many US states, indicating wide geographic coverage of air quality monitoring.
- Higher concentrations of AQI observations appear in densely populated and industrial regions.
- The visualization helps identify areas with more frequent air quality activity and pollution monitoring.

Which top 10 counties recorded the highest average AQI values?

VISUALIZATION

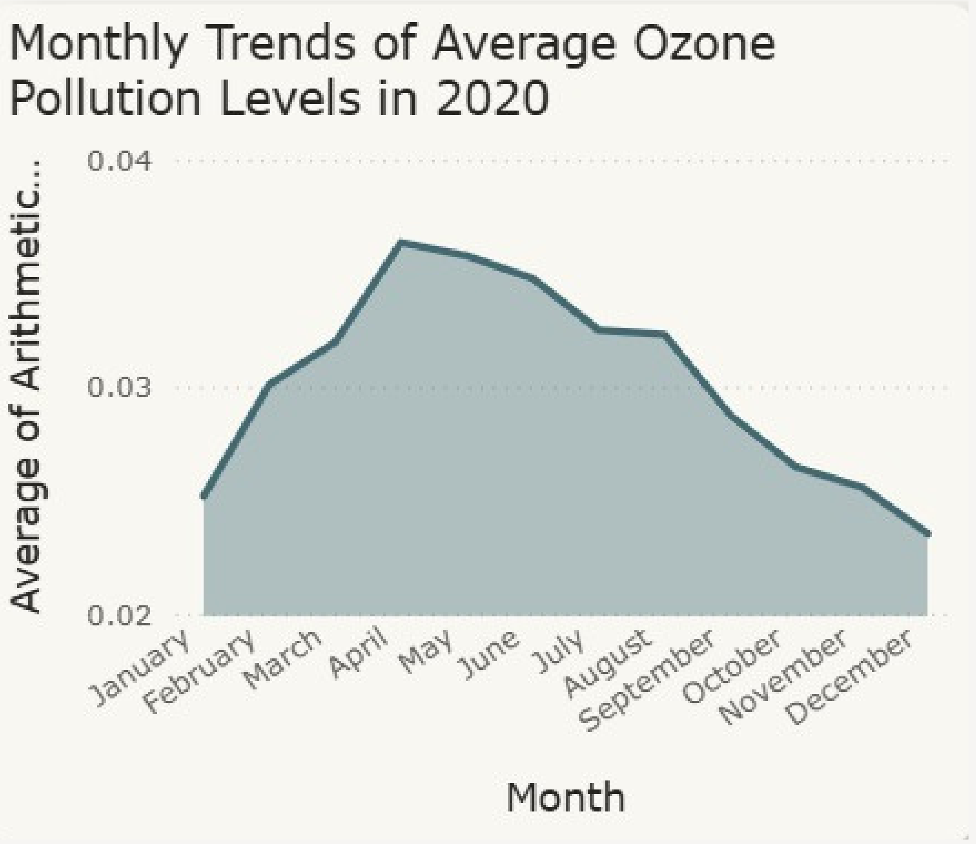


RESULTS

- San Bernardino County recorded the highest average AQI among the top 10 counties.
- Most counties had AQI values above 50, which reflects moderate air quality conditions.
- The visualization reveals clear variations in pollution levels between counties, suggesting that some areas experience worse air quality than others.

How did ozone pollution levels change across the months of 2020?

VISUALIZATION

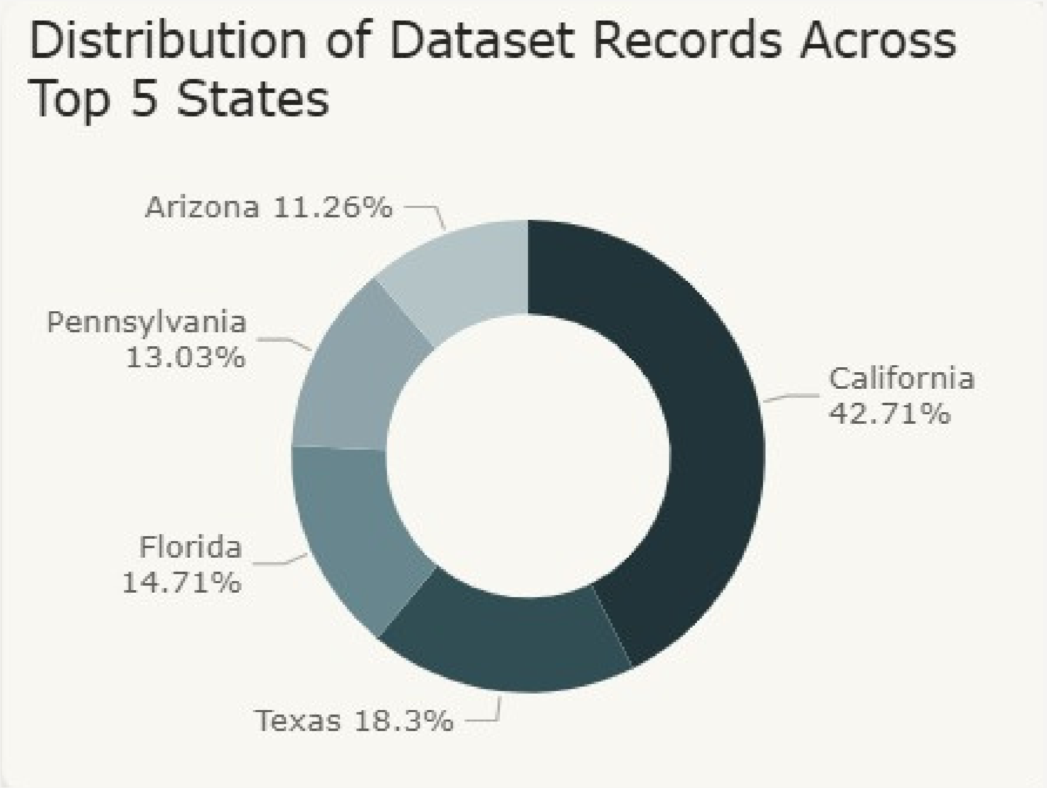


RESULTS

- Ozone pollution levels increased steadily during the first half of 2020.
- Peak ozone concentration was observed around May and June.
- Pollution levels gradually declined toward the end of the year.

Which states contributed the highest number of records in the dataset?

VISUALIZATION

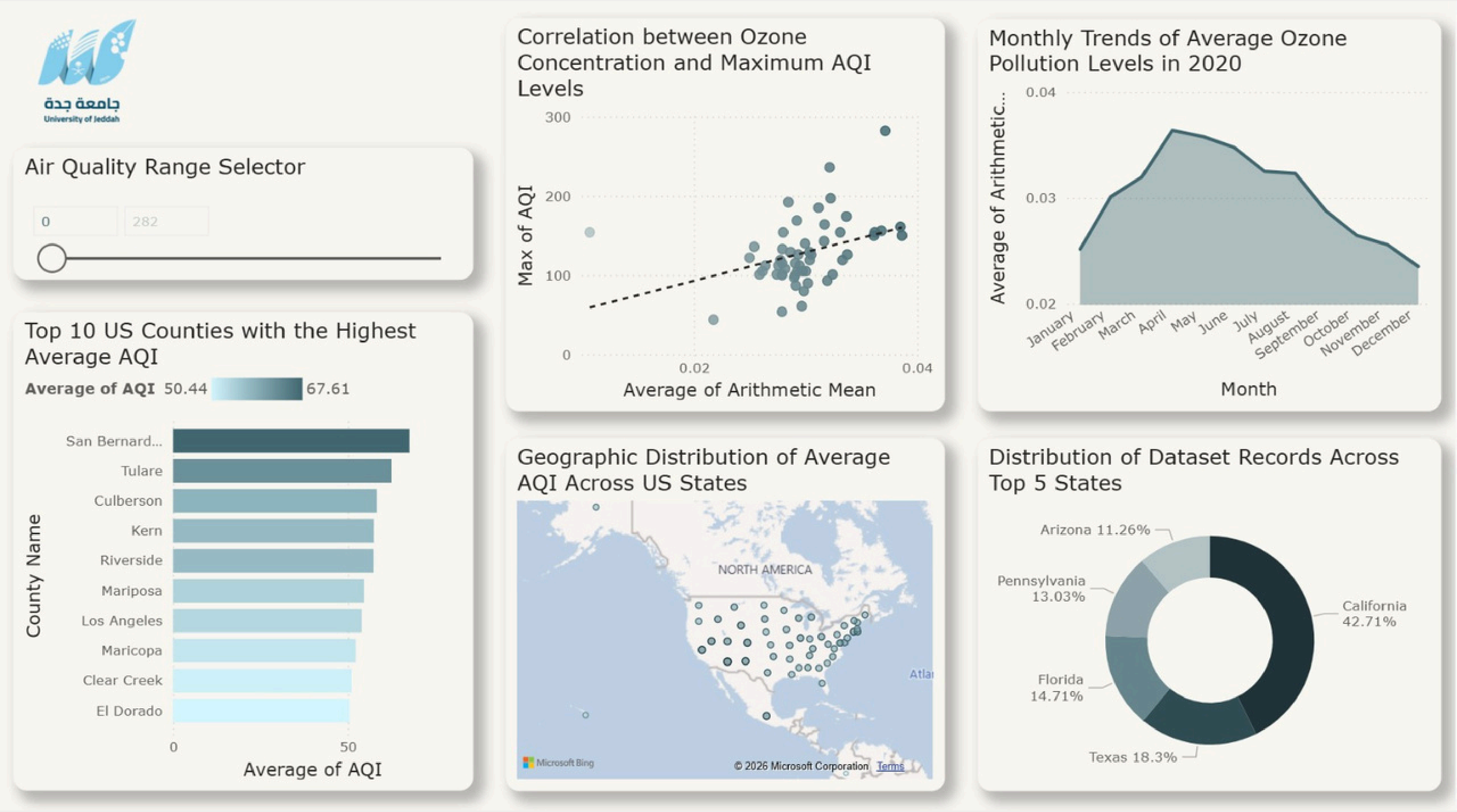


RESULTS

- California contributed the largest share of dataset records.
- Texas and Florida also represented significant portions of the dataset.
- The dataset records are unevenly distributed across states.

Dashboard

POWER BI DASHBOARD



THANKYOU

ANY QUESTIONS?